

Cross-mission expiMap analysis recovers established tissue responses and identifies complementary pathway shifts in mouse spaceflight transcriptomes

Jason Trinh | NASA Space Life Sciences Training Program, NASA Ames Research Center

jasontrinh@berkeley.edu

github.com/jasont314/nasa-mouse

TAKE-HOME

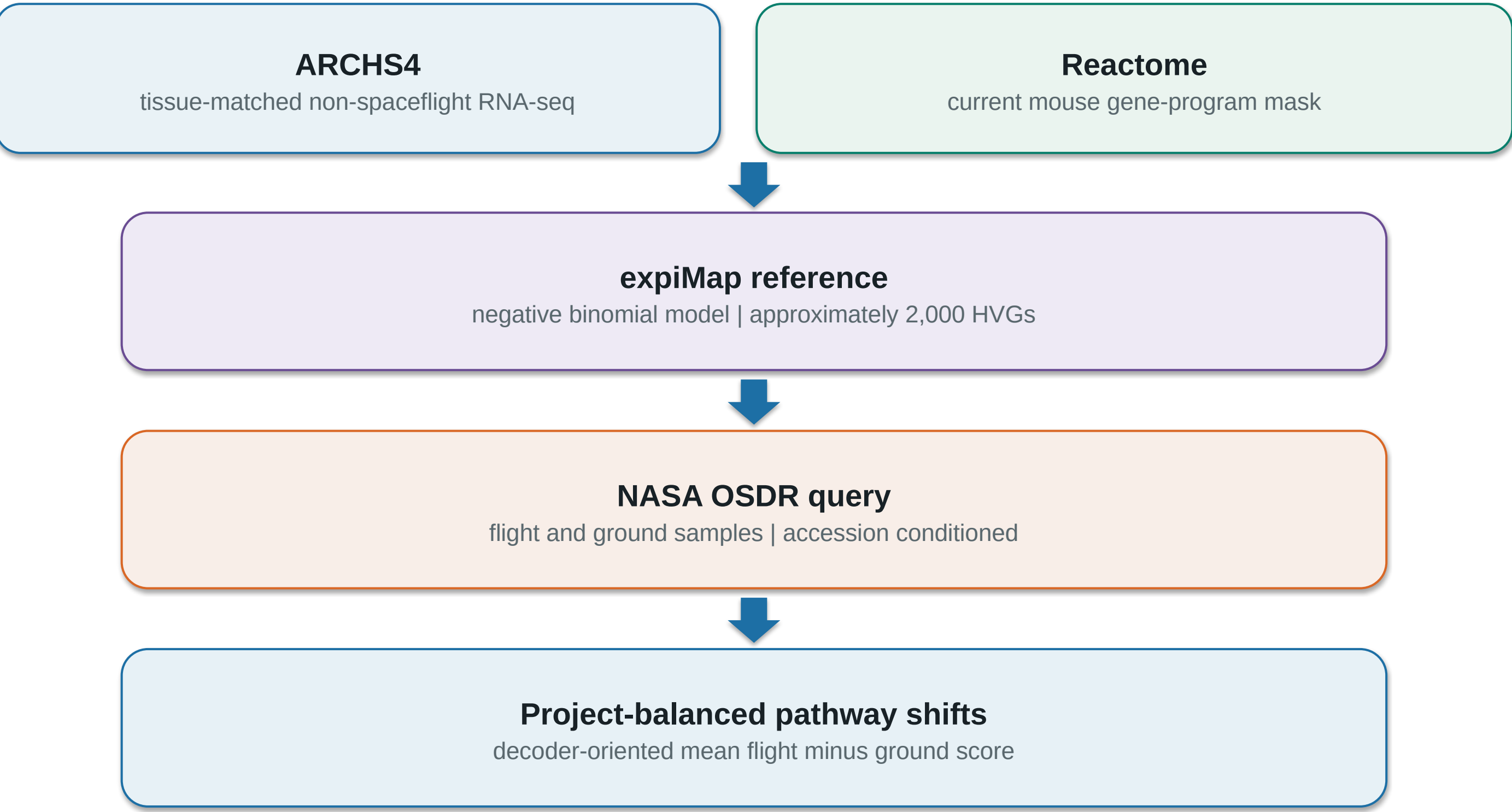
Cross-mission pathway mapping recovered established tissue responses and prioritized complementary maintenance and immune hypotheses; spleen showed the strongest multi-method evidence.

QUESTION & APPROACH

Can a tissue-matched, pathway-constrained reference distinguish recurring flight-associated programs from mission-specific variation?

Public mouse RNA-seq studies differ in mission, strain, sex, duration, hardware, collection endpoint, and sequencing protocol.

REFERENCE-MAPPING WORKFLOW



PRIMARY MODEL SCOPE

Tissue	ARCHS4	OSDR	Projects	Programs
Thymus	1,362	117	5	387
Skin	2,593	151	4	319
Liver	5,000	197	9	364
Spleen	6,289	100	5	360

OSDR counts are primary analysis samples; spleen excludes a condition-strain-confounded project.

ROBUSTNESS GATE

ssGSEA	Preranked GSEA	Held-out projects
3 full trainings	Composition proxies	Member genes

Pathways were interpreted after project direction, complete retraining, conventional enrichment, composition sensitivity, and member-gene review.

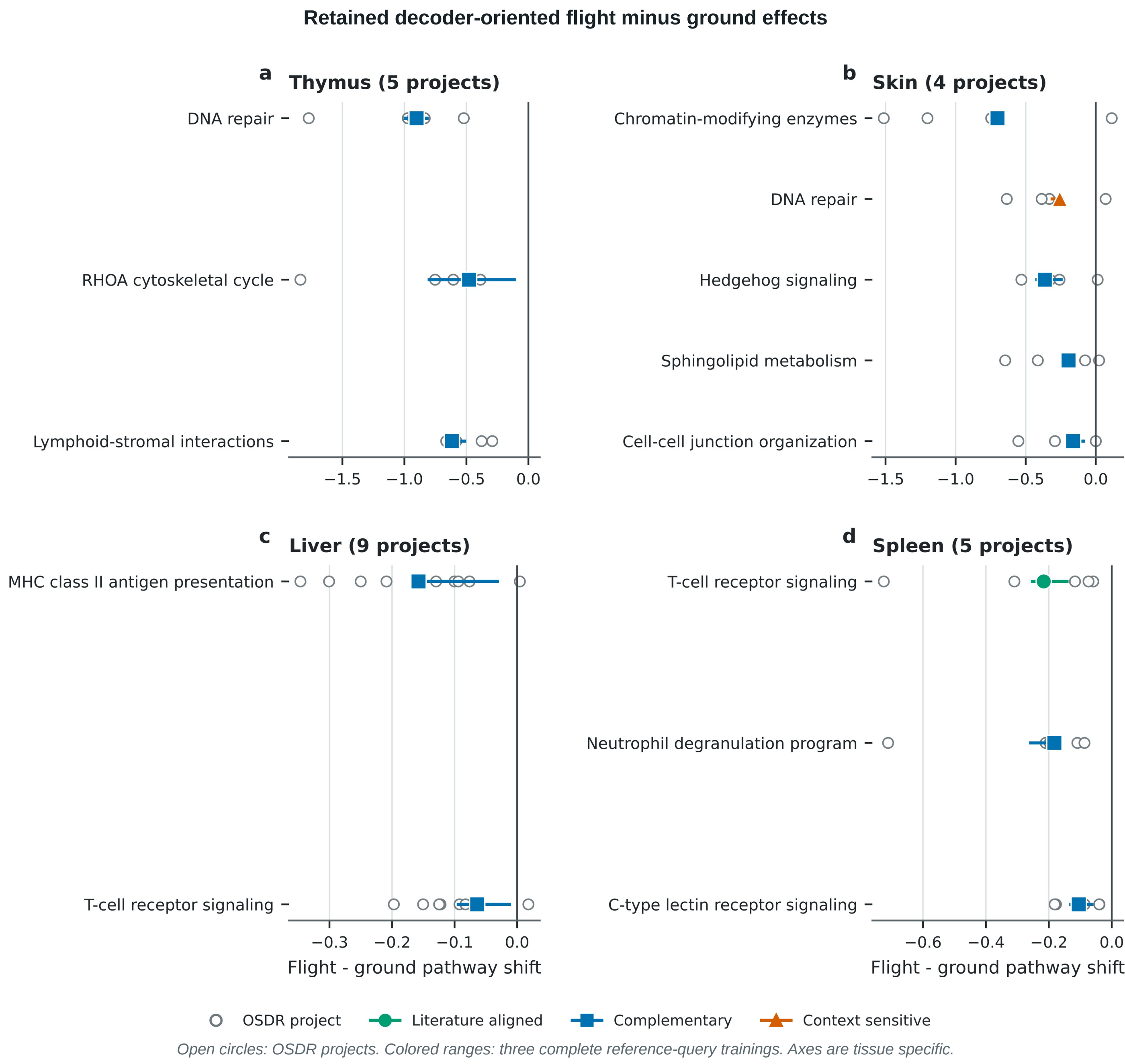
TRAINING 3 x 300 hidden units | reference <=400 epochs | query 250 epochs | NVIDIA A100

Complementary programs are annotated Reactome nodes that extend prior literature; de novo nodes were not retained in the final models.

ANALYSIS SCOPE

Kidney remained exploratory. Soleus, lung, and retina were not advanced to the primary biological narrative.

PATHWAY SHIFTS

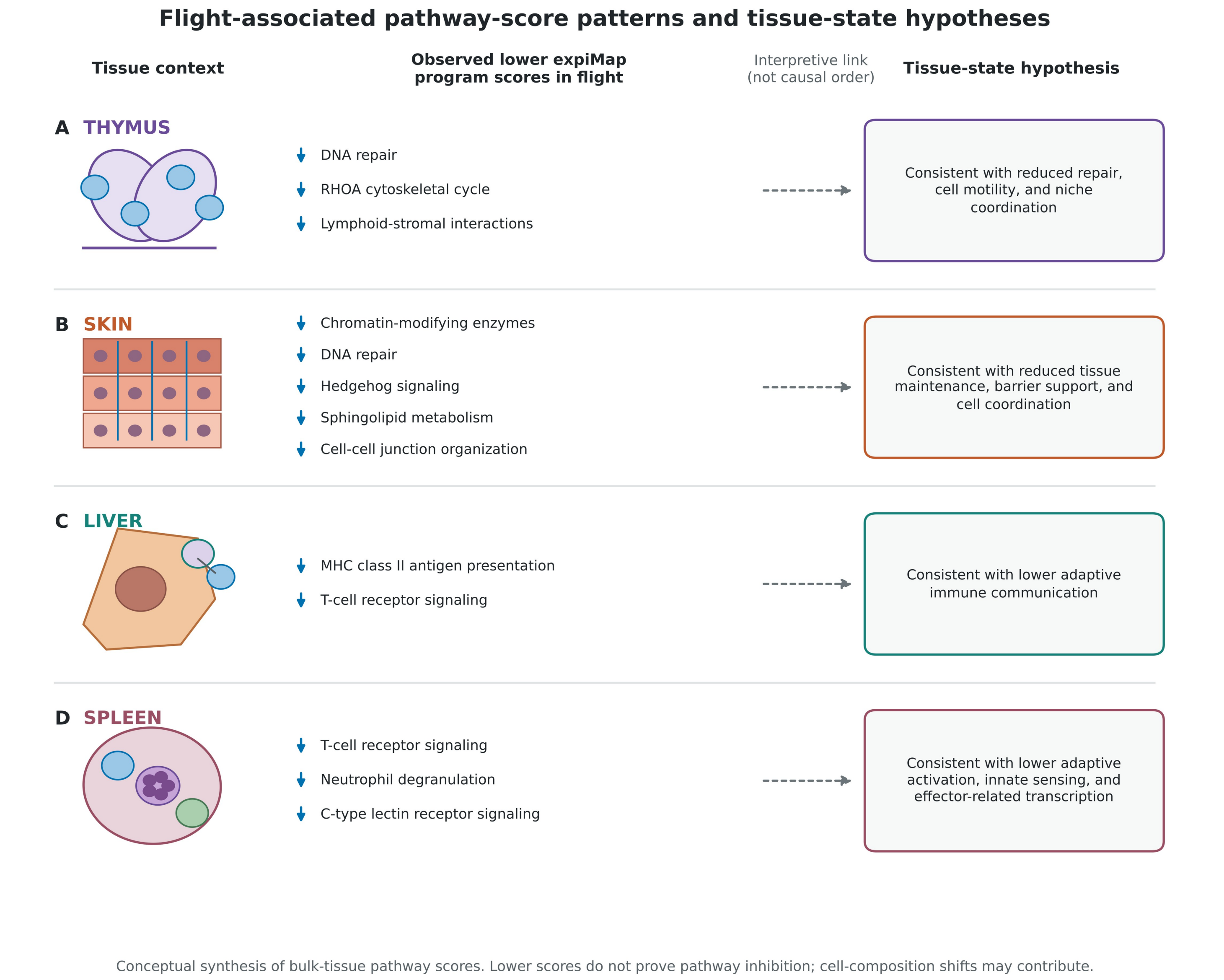


EVIDENCE AT A GLANCE

THYMUS	Lower DNA repair and cytoskeletal scores; expiMap also suggests lower niche interaction, but GSEA does not.	5/5 projects 3/3 trainings FDR <0.001 / 0.154 / 1.000
SKIN	Lower regulation, repair, Hedgehog, sphingolipid, and junction scores.	3/4 projects 3/3 trainings 3 of 5 FDR <0.05
LIVER	Lower MHC class II antigen presentation and T-cell receptor scores amid mixed metabolism.	8/9 projects 3/3 trainings FDR 0.121 / 0.051
SPLEEN	Lower T-cell receptor, degranulation, and C-type lectin receptor programs.	5/5 projects 3/3 trainings all FDR <0.05

FDR values are from preranked GSEA. Directional robustness and FDR are complementary evidence, not interchangeable thresholds.

BIOLOGICAL INTERPRETATION



WHAT THIS ADDS TO PRIOR LITERATURE

- Thymus:** Repair and cytoskeletal reduction adds a possible niche-coordination layer to known involution.
- Skin:** Multiple programs converge on lower tissue maintenance and barrier coordination.
- Liver:** A lower immune-communication axis complements the better-known metabolic response.
- Spleen:** Strongest evidence: adaptive activation and innate sensing and effector transcription decrease together.

INTERPRETATION AND NEXT TESTS

- Lower means a lower decoder-oriented latent score, not biochemical inhibition.
- Bulk tissue cannot distinguish cell abundance from altered cell state.
- Validate with independent single-cell, spatial, biochemical, and functional studies.

CONCLUSION

The cross-mission design narrows broad tissue phenotypes into testable pathway-level hypotheses without claiming causality or cell-type resolution.

SELECTED REFERENCES

1 Gebre et al., Nucleic Acids Res 2025 (OSDR). 2 Lotfollahi et al., Nat Cell Biol 2023 (expiMap). 3 Lotfollahi et al., Nat Biotechnol 2022 (scArches). 4 Lachmann et al., Nat Commun 2018 (ARCHS4). 5 Milacic et al., Nucleic Acids Res 2024 (Reactome). 6 Horie et al., Sci Rep 2019 (thymus). 7 Cope et al., Commun Med 2024 (skin). 8 Beheshti et al., Sci Rep 2019 (liver). 9 Gridley et al., J Appl Physiol 2009 (spleen).

DATA, CODE, AND CONTACT

NASA OSDR API | ARCHS4 | Reactome
github.com/jasont314/nasa-mouse
jasontrinh@berkeley.edu

NASA/SLSTP

NASA Ames Research Center
Moffett Field, California